

Lecture 10: Fundamentals of Topologies

Network Topology

- **Definition:** The physical layout or arrangement of how computers are connected in a network.
 - **Importance:** It shows how devices communicate and is a key design factor in networking.
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Bus Topology

- **Layout:** All computers and devices are connected to a single central cable (called the bus).
 - **Advantages:** Simple design, low cost, easy to expand.
 - **Disadvantages:** If the main cable breaks, the whole network stops working.
 - **Example:** Small office networks in earlier times.
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Ring Topology

- **Layout:** Computers are connected in a circular path.
 - **Working:** Data travels in one direction around the ring, passing through each computer until it reaches the destination.
 - **Advantages:** Equal access for all devices.
 - **Disadvantages:** If one computer or link fails, the entire network can stop.
 - **Example:** Token Ring networks used in older LANs.
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Star Topology

- **Layout:** All computers connect to a central device (hub or switch).
- **Working:** To send data, the sender passes it to the hub, which forwards it to the destination.
- **Advantages:** Easy to set up, expand, and troubleshoot. If one link fails, only that computer is affected.
- **Disadvantages:** If the hub fails, the whole network stops.
- **Example:** Modern LANs in homes and offices.

Comparison of Network Topologies

Feature	Bus Topology	Ring Topology	Star Topology
Layout	All devices connected to a single central cable (bus).	Devices connected in a circular loop.	All devices connected to a central hub/switch.
Data Flow	Data travels along the bus; all devices share the same cable.	Data travels in one direction around the ring.	Data goes to hub, then forwarded to destination.
Advantages	Simple, low cost, easy to expand.	Equal access for all devices.	Easy to set up, expand, and troubleshoot.
Disadvantages	If main cable fails, whole network stops.	If one device/link fails, entire network stops.	If hub fails, whole network stops.
Example Use	Small office networks (older LANs).	Token Ring networks (older LANs).	Modern LANs in homes and offices.

MCQs on Fundamentals of Topologies

1.

The physical layout of how computers are connected in a network is called:

- A) Protocol
 - B) Topology
 - C) Bandwidth
 - D) Router
-

2.

Which topology uses a single central cable as the backbone?

- A) Ring
 - B) Star
 - C) Bus
 - D) Mesh
-

3.

In bus topology, if the main cable fails:

- A) Only one computer is affected
 - B) The entire network stops working
 - C) The hub continues to work
 - D) Nothing happens
-

4.

Which topology connects computers in a circular path?

- A) Bus
 - B) Ring
 - C) Star
 - D) Mesh
-

5.

In ring topology, data travels:

- A) In both directions
 - B) In one direction around the loop
 - C) Randomly
 - D) Through the hub
-

6.

Which topology uses a central device like a hub or switch?

- A) Bus
 - B) Ring
 - C) Star
 - D) Mesh
-

7.

In star topology, if one link fails:

- A) The whole network fails
 - B) Only that computer is affected
 - C) All computers stop working
 - D) The hub shuts down
-

8.

Which topology is easiest to expand and troubleshoot?

- A) Bus
 - B) Ring
 - C) Star
 - D) Mesh
-

9.

Which topology is cheapest to set up?

- A) Bus
 - B) Ring
 - C) Star
 - D) Mesh
-

10.

Which topology is most commonly used in modern LANs?

- A) Bus
 - B) Ring
 - C) Star
 - D) Mesh
-

11.

In bus topology, communication happens through:

- A) Hub
 - B) Switch
 - C) Central cable (bus)
 - D) Router
-

12.

In ring topology, each computer acts as:

- A) A hub
 - B) A repeater
 - C) A server
 - D) A router
-

13.

Which topology allows equal access for all devices?

- A) Bus
 - B) Ring
 - C) Star
 - D) Mesh
-

14.

Which topology requires the least amount of cable?

- A) Bus
- B) Ring
- C) Star
- D) Mesh

15.

Which topology depends heavily on a central device?

- A) Bus
 - B) Ring
 - C) Star
 - D) Mesh
-

16.

Which topology is affected most by a single cable failure?

- A) Bus
 - B) Ring
 - C) Star
 - D) Mesh
-

17.

Which topology is affected most by a hub failure?

- A) Bus
 - B) Ring
 - C) Star
 - D) Mesh
-

18.

Which topology is simple but not reliable?

- A) Bus
 - B) Ring
 - C) Star
 - D) Mesh
-

19.

Which topology is reliable but more expensive?

- A) Bus
- B) Ring

- C) Star
 - D) Mesh
-

20.

Topology is a major consideration in:

- A) Computer networking design
 - B) Software development
 - C) Database management
 - D) Operating systems
-

✓ Answers with Explanations

1. **B – Topology** → Defines physical layout of network.
 2. **C – Bus** → Uses one central cable.
 3. **B – Entire network stops** → Bus cable failure halts communication.
 4. **B – Ring** → Devices connected in a circle.
 5. **B – One direction** → Data flows around the loop.
 6. **C – Star** → Uses hub/switch.
 7. **B – Only that computer** → Star isolates failures.
 8. **C – Star** → Easy to expand and troubleshoot.
 9. **A – Bus** → Cheapest due to single cable.
 10. **C – Star** → Most common in modern LANs.
 11. **C – Central cable** → Bus topology backbone.
 12. **B – Repeater** → Each device forwards data.
 13. **B – Ring** → Equal access for all.
 14. **A – Bus** → Requires least cable.
 15. **C – Star** → Depends on hub/switch.
 16. **A – Bus** → Cable failure stops all.
 17. **C – Star** → Hub failure halts network.
 18. **A – Bus** → Simple but fragile.
 19. **C – Star** → Reliable but costlier.
 20. **A – Networking design** → Topology is key in network planning.
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Lecture 10: Network Architecture, OSI Model, TCP/IP Model, and Network Addressing

Network Architecture

- **Definition:** The design or framework of a computer network.
- It explains how physical components are arranged, how they work together, and which communication rules (protocols) are used.
- Example: OSI and TCP/IP layered architectures.

OSI Model (ISO's Open Systems Interconnection)

- A **conceptual model** developed by ISO to standardize communication between different systems.
- Divides communication into **7 layers**:
 1. **Physical:** Deals with cables, signals, and hardware.
 2. **Data Link:** Ensures error-free transmission using physical addresses (MAC).
 3. **Network:** Chooses best path, uses logical addresses (IP).
 4. **Transport:** Ensures reliable delivery, error control, flow control.
 5. **Session:** Manages sessions (start, maintain, end connections).
 6. **Presentation:** Converts data formats, handles encryption and compression.
 7. **Application:** Provides services like email, file transfer, web browsing.

TCP/IP Model

- A **suite of protocols** used on the internet.
- Provides rules for breaking data into packets, addressing, transmitting, and receiving.
- Has **4 layers** (mapped to OSI):
 1. **Application Layer:** Uses protocols like HTTP, FTP; interacts with apps.
 2. **Transport Layer:** Uses TCP/UDP; ensures reliable delivery.

- 3. **Internet Layer:** Handles logical addressing and routing (IP).
 - 4. **Network Access Layer:** Deals with physical transmission (hardware, cables).
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Network Addressing

- **Definition:** Like house addresses, network addresses uniquely identify devices.
 - Two types:
 - **Logical Address:** IP address (used for routing).
 - **Physical Address:** MAC address (unique hardware ID).
 - Example: Your computer's IP address helps data reach it, while the MAC address identifies the exact device.
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Key Takeaway

- **Network architecture** is the overall design.
 - **OSI model** has 7 layers, each with specific functions.
 - **TCP/IP model** is practical, with 4 layers used on the internet.
 - **Network addressing** ensures every device is uniquely identified, just like houses in a city.
-

MCQs – Network Architecture

1.

Network architecture refers to:

- A) Software design only
 - B) Framework of computer network design
 - C) Database structure
 - D) Operating system layout
-

2.

Which model was developed by ISO to standardize communication?

- A) TCP/IP
 - B) OSI Model
 - C) HTTP Model
 - D) Ethernet
-

3.

How many layers are in the OSI model?

- A) 4
 - B) 5
 - C) 6
 - D) 7
-

4.

Which OSI layer deals with cables, signals, and hardware?

- A) Data Link
 - B) Physical
 - C) Network
 - D) Transport
-

5.

Which OSI layer ensures error-free transmission using MAC addresses?

- A) Network

- B) Data Link
 - C) Transport
 - D) Session
-

6.

Which OSI layer selects the best path and uses IP addresses?

- A) Network
 - B) Transport
 - C) Data Link
 - D) Physical
-

7.

Which OSI layer ensures reliable delivery, error control, and flow control?

- A) Transport
 - B) Session
 - C) Presentation
 - D) Application
-

8.

Which OSI layer manages sessions between applications?

- A) Session
 - B) Transport
 - C) Network
 - D) Physical
-

9.

Which OSI layer handles encryption, compression, and data format conversion?

- A) Presentation
 - B) Application
 - C) Transport
 - D) Network
-

10.

Which OSI layer provides services like email and file transfer?

- A) Application
 - B) Presentation
 - C) Transport
 - D) Session
-

11.

TCP/IP model has how many layers?

- A) 7
 - B) 5
 - C) 4
 - D) 6
-

12.

Which TCP/IP layer uses protocols like HTTP and FTP?

- A) Application
 - B) Transport
 - C) Internet
 - D) Network Access
-

13.

Which TCP/IP layer uses TCP/UDP for reliable delivery?

- A) Transport
 - B) Application
 - C) Internet
 - D) Network Access
-

14.

Which TCP/IP layer handles logical addressing and routing?

- A) Internet
- B) Transport
- C) Application
- D) Physical

15.

Which TCP/IP layer deals with hardware and physical transmission?

- A) Network Access
 - B) Internet
 - C) Transport
 - D) Application
-

16.

Network addressing is similar to:

- A) Telephone numbers
 - B) House addresses
 - C) Passwords
 - D) File names
-

17.

Which address uniquely identifies hardware devices?

- A) IP address
 - B) MAC address
 - C) Domain name
 - D) URL
-

18.

Which address is logical and used for routing?

- A) MAC address
 - B) IP address
 - C) Serial number
 - D) Port number
-

19.

Which OSI layer converts logical addresses to physical addresses?

- A) Network
- B) Data Link

- C) Transport
 - D) Session
-

20.

Which TCP/IP layer is responsible for packet forwarding?

- A) Network Access
 - B) Internet
 - C) Transport
 - D) Application
-

✓ Answers with Explanations

1. **B – Framework of computer network design** → Network architecture defines layout, components, and protocols.
2. **B – OSI Model** → Developed by ISO for standardization.
3. **D – 7 layers** → OSI has 7 abstraction layers.
4. **B – Physical** → Deals with cables, signals, hardware.
5. **B – Data Link** → Ensures error-free transmission using MAC addresses.
6. **A – Network** → Selects best path, uses IP addresses.
7. **A – Transport** → Ensures reliable delivery, error/flow control.
8. **A – Session** → Manages sessions between applications.
9. **A – Presentation** → Handles encryption, compression, format conversion.
10. **A – Application** → Provides services like email, file transfer.
11. **C – 4 layers** → TCP/IP has 4 layers.
12. **A – Application** → Uses HTTP, FTP protocols.
13. **A – Transport** → Uses TCP/UDP for reliable delivery.
14. **A – Internet** → Handles logical addressing and routing.
15. **A – Network Access** → Deals with hardware and physical transmission.
16. **B – House addresses** → Network addresses are unique like house addresses.
17. **B – MAC address** → Identifies hardware devices uniquely.
18. **B – IP address** → Logical address used for routing.
19. **A – Network** → Converts logical (IP) to physical (MAC).
20. **A – Network Access** → Responsible for packet forwarding physically.

Lecture: Network Addressing

- A **network address** is like a house address.
 - It must be **unique** so that data reaches the correct device without confusion.
 - Every device on a network (computer, printer, smartphone, tablet) needs an address to communicate.
 - Addresses can be **logical** (IP address) or **physical** (MAC address).
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◆ Physical Address (MAC Address)

- Built into the hardware (Network Interface Card).
 - Permanent and globally unique.
 - 48-bit number.
 - Example: Like a serial number engraved on a device.
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◆ Logical Address (IP Address)

- Assigned by the Internet Service Provider (ISP) or network administrator.
 - Temporary and can be changed.
 - 32-bit number in IPv4.
 - Example: 192.168.108.105
 - Divided into two parts:
 - **Network portion** → identifies the network.
 - **Host portion** → identifies the specific device.
-

◆ IPv4 Address

- Stands for **Internet Protocol version 4**.
 - Uses 32 binary bits, written as four numbers separated by dots.
 - Example: 192.168.1.1
 - Works like a phone number or car registration — shows ownership and location.
-

Data Communication Basics

- **Message, sender, receiver, medium, protocol** are the five components.
 - Signals can be **analog** (continuous waves) or **digital** (bits).
 - **Data rate**: speed of data transfer (bits per second).
 - **Baud rate**: number of signals per second.
 - Transmission media:
 - **Guided** (cables like fiber optics, copper wires).
 - **Unguided** (wireless like radio waves, Wi-Fi).
-

Key Takeaway

- **Physical address (MAC)**: permanent hardware ID.
- **Logical address (IP)**: temporary, changeable, used for routing.
- **IPv4**: 32-bit logical address, essential for internet communication.
- Network addressing ensures every device can be uniquely identified and reached, just like houses in a city.

MCQs – Network Addressing & Communication

1.

A network address is similar to:

- A) A password
 - B) A house address
 - C) A file name
 - D) A phone contact
-

2.

Which type of address is permanent and built into hardware?

- A) Logical address
 - B) Physical address (MAC)
 - C) IP address
 - D) Domain name
-

3.

Which type of address is temporary and can be changed?

- A) Physical address
 - B) Logical address (IP)
 - C) MAC address
 - D) Serial number
-

4.

A physical address is usually:

- A) 32-bit
 - B) 48-bit
 - C) 64-bit
 - D) 16-bit
-

5.

A logical IPv4 address is:

- A) 32-bit

- B) 48-bit
 - C) 64-bit
 - D) 128-bit
-

6.

Which organization assigns IP addresses to devices on the internet?

- A) ISP or network administrator
 - B) Hardware manufacturer
 - C) Operating system vendor
 - D) Database manager
-

7.

An IPv4 address is divided into:

- A) Host and network portions
 - B) User and password portions
 - C) Sender and receiver portions
 - D) Hardware and software portions
-

8.

Example of an IPv4 address is:

- A) 192.168.108.105
 - B) 00:1A:2B:3C:4D:5E
 - C) www.google.com
 - D) 123456789
-

9.

Which address uniquely identifies a device's hardware?

- A) IP address
 - B) MAC address
 - C) Domain name
 - D) Port number
-

10.

Which address is provided by ISP and can be changed?

- A) MAC address
 - B) IP address
 - C) Serial number
 - D) Hardware ID
-

11.

Data communication refers to:

- A) Exchange of digital messages between devices
 - B) Writing a letter
 - C) Printing documents
 - D) Manual file transfer
-

12.

Analog signals are:

- A) Continuous waves that vary with time
 - B) Discrete values in bits
 - C) Fixed numbers
 - D) Digital codes only
-

13.

Digital signals represent data as:

- A) Continuous waves
 - B) Bits (0s and 1s)
 - C) Colors
 - D) Sounds only
-

14.

Data rate is measured in:

- A) Bytes per minute
 - B) Bits per second
 - C) Hertz
 - D) Kilograms
-

15.

Baud rate refers to:

- A) Number of signals per second
 - B) Number of bits per second
 - C) Number of users connected
 - D) Number of devices in a network
-

16.

A communication system has how many basic components?

- A) 3
 - B) 4
 - C) 5
 - D) 6
-

17.

Which of the following is NOT a component of communication system?

- A) Message
 - B) Sender
 - C) Receiver
 - D) Password
-

18.

Effectiveness of communication depends on:

- A) Delivery, accuracy, timeliness
 - B) Passwords and usernames
 - C) Hardware only
 - D) Software only
-

19.

Guided transmission media includes:

- A) Fiber optic cables
- B) Radio waves

- C) Wi-Fi signals
 - D) Satellite links
-

20.

Unguided transmission media includes:

- A) Copper wires
 - B) Fiber optics
 - C) Wireless signals (radio, Wi-Fi)
 - D) Ethernet cables
-

✓ **Answers with Explanations**

1. **B – House address** → Network address uniquely identifies devices like house addresses.
 2. **B – Physical address (MAC)** → Permanent, built into NIC hardware.
 3. **B – Logical address (IP)** → Temporary, can be changed.
 4. **B – 48-bit** → MAC address length.
 5. **A – 32-bit** → IPv4 address length.
 6. **A – ISP or network administrator** → Assigns logical IP addresses.
 7. **A – Host and network portions** → IPv4 divided into two parts.
 8. **A – 192.168.108.105** → Example of IPv4.
 9. **B – MAC address** → Identifies hardware uniquely.
 10. **B – IP address** → Provided by ISP, changeable.
 11. **A – Exchange of digital messages** → Definition of data communication.
 12. **A – Continuous waves** → Analog signals vary with time.
 13. **B – Bits (0s and 1s)** → Digital signals represent discrete values.
 14. **B – Bits per second** → Unit of data rate.
 15. **A – Number of signals per second** → Baud rate definition.
 16. **C – 5** → Components: message, sender, receiver, medium, protocol.
 17. **D – Password** → Not a communication component.
 18. **A – Delivery, accuracy, timeliness** → Key characteristics of effective communication.
 19. **A – Fiber optic cables** → Guided media uses physical paths.
 20. **C – Wireless signals (radio, Wi-Fi)** → Unguided media uses air as medium.
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